

1. Scope and Objective

Thiolate-protected noble metal nanoclusters combine an atomically precise metal kernel with a dense organic shell. Any catalytic or sensing function that requires direct contact between a substrate molecule and a metal atom is therefore governed not only by the electronic structure of the kernel but also by whether the substrate can physically reach a metal site. When such clusters are assembled into an ordered two-dimensional arrangement, neighbouring clusters may occlude further surface area, and the accessible fraction of the metal may be reduced relative to the isolated cluster.

The present work quantifies that reduction for an Ag₄₄(SR)₃₀ cluster. Two geometric descriptors of site accessibility are defined and evaluated: the accessible solid-angle fraction and the atomic solvent-accessible surface area. Both are computed for every silver atom, first in the isolated cluster, then in a two-dimensional hexagonal layer in which clusters occupy the six vertices of a regular hexagon together with its centre, and finally in three-dimensional superlattices obtained by stacking such layers. The ratio of assembly to isolated-cluster value, denoted the retention, isolates the contribution of interparticle packing to metal burial. A central outcome is that the two geometries behave entirely differently, so the dimensionality of the assembly, rather than its in-plane density, governs the result.

2. Structural Model

The input geometry comprises 554 atoms with the composition Ag₄₄S₃₀C₂₁₀N₃₀₀O₁₂₀H₁₂₀, corresponding to thirty thiolate ligands of stoichiometry C₇H₄NO₄S per ligand. The stoichiometry is consistent with a nitro-substituted mercaptobenzoate ligand, and the ligand identity should be confirmed against the synthetic source before publication.

Radial analysis about the centre of mass reproduces the established three-shell architecture of the Ag₄₄(SR)₃₀ family. Twelve silver atoms lie between 2.29 and 3.13 Angstrom and constitute the icosahedral kernel. Twenty silver atoms lie between 4.13 and 4.93 Angstrom and form the dodecahedral shell. The remaining twelve silver atoms lie between 6.28 and 7.08 Angstrom and belong to the six Ag₂(SR)₅ mount units that terminate the particle. The outermost atom of the cluster, a ligand oxygen, lies 14.36 Angstrom from the centre of mass.

3. Theory

3.1 Hard-sphere representation

Each atom i is represented by a sphere of Bondi van der Waals radius R_i centred at position r_i . A probe molecule is represented by a sphere of radius r_p . The probe is treated as rigid and structureless, so that its centre may occupy any point whose distance from every atom centre satisfies the non-overlap condition below.

$$|x - r_j| \geq R_j + r_p \quad \text{for all } j \quad (1)$$

The set of admissible probe-centre positions defined by Equation 1 is the complement of a union of spheres of inflated radius $R_j + r_p$. All accessibility measures below are functionals of that set.

3.2 Accessible solid-angle fraction

Consider a metal atom M at position r_M . A probe approaching along the unit direction u follows the straight path

$$x(t) = r_M + t u, \quad t \geq t_0 = R_M + r_p \quad (2)$$

The lower limit t_0 corresponds to the probe in van der Waals contact with the metal atom, which is the closest approach permitted by Equation 1 and therefore the terminal point of a successful approach.

Writing $d_j = r_j - r_M$, $s_j = u \cdot d_j$ and $D_j^2 = |d_j|^2$, the squared distance from the probe centre to obstacle j along the path is

$$f_j(t) = |x(t) - r_j|^2 = t^2 - 2s_j t + D_j^2 \quad (3)$$

The quadratic f_j is minimised at $t = s_j$. Restricting the minimisation to the physically accessible interval $t \geq t_0$ gives the closed-form minimum

$$f_j^{\min} = D_j^2 - s_j^2 \quad \text{if } s_j \geq t_0, \quad f_j^{\min} = t_0^2 - 2t_0 s_j + D_j^2 \quad \text{otherwise} \quad (4)$$

The second branch of Equation 4 is essential and is the point at which a naive ray-casting implementation fails. Ligand atoms directly bonded to the metal frequently satisfy $D_j < R_j + r_p$, so a ray traced from the nucleus would report every direction as blocked and would assign zero accessibility to atoms that are in fact exposed. Anchoring the path at the contact radius t_0 removes the artefact without introducing an adjustable parameter.

The direction u is defined as open when no obstacle intersects the path, that is

$$f_j^{\min} \geq (R_j + r_p)^2 \quad \text{for all } j \neq M \quad (5)$$

The accessible solid-angle fraction of atom M is the measure of the open set of directions normalised by the full sphere,

$$\Omega_M / 4\pi = (1 / 4\pi) \int_{S^2} \chi(u) d\Omega \approx (1 / N) \sum_k \chi(u_k) \quad (6)$$

where χ is the indicator function of Equation 5. The descriptor is dimensionless and bounded between zero and unity. A bare atom in vacuum attains unity, and an atom for which no straight-line approach exists attains exactly zero. Quadrature points u_k are generated by the Fibonacci construction

$$\cos \theta_k = 1 - (2k + 1) / N, \quad \phi_k = \pi (1 + \sqrt{5}) k, \quad k = 0, \dots, N - 1 \quad (7)$$

which distributes points with low discrepancy and converges faster than pseudorandom sampling for smooth indicator sets.

3.3 Atomic solvent-accessible surface area

The second descriptor is the Shrake and Rupley solvent-accessible surface area of the metal atom taken in isolation. Points are distributed on the sphere of radius $R_M + r_p$ about the metal, and the area is the unburied fraction scaled by the full sphere area,

$$A_M = 4\pi (R_M + r_p)^2 \times (N_{\text{free}} / N) \quad (8)$$

A point is buried when it lies within $R_j + r_p$ of any other atom j .

The two descriptors are complementary. Equation 6 measures whether a straight-line approach corridor exists and is the quantity relevant to substrate delivery. Equation 8 measures contact area and is directly comparable with values reported for other clusters. A site may possess appreciable area while having a narrow approach corridor, so both quantities are reported.

3.4 Construction of the hexagonal assembly

Clusters placed at the six vertices of a regular hexagon together with its centre generate, on periodic repetition, the two-dimensional triangular Bravais lattice. The hexagon edge length equals the centre-

to-vertex distance and equals the nearest-neighbour separation a . Taking the assembly plane perpendicular to the unit normal n , with orthonormal in-plane vectors e_1 and e_2 , the lattice vectors are

$$a_1 = a e_1, \quad a_2 = a (e_1 / 2 + (\sqrt{3} / 2) e_2) \quad (9)$$

Each lattice point carries one complete cluster in a fixed orientation. In the calculations reported below the central cluster is surrounded by a five by five block of images, so that the two nearest coordination shells of neighbours are represented in full. Increasing the block to seven by seven changes no reported quantity, which confirms that the truncation is converged.

Three-dimensional superlattices are generated by translating the layer along the normal with interlayer separation c , with an in-plane offset that repeats after m layers,

$$a_3 = c n, \quad \Delta_k = (k \bmod m) \times (a_1 + a_2) / 3 \quad (10)$$

with $m = 1$ for simple hexagonal stacking (AAA), $m = 2$ for hexagonal close packed stacking (ABAB) and $m = 3$ for cubic close packed stacking (ABCABC). The offset vector places a cluster of one layer over the threefold hollow formed by three clusters of the layer below.

The retention is defined as the ratio of an accessibility descriptor evaluated in the assembly to the same descriptor evaluated for the isolated cluster,

$$\mathcal{R}(\Omega) = \Sigma_M \Omega_M^{\text{sheet}} / \Sigma_M \Omega_M^{\text{cluster}} \quad (11)$$

Retention equal to unity indicates that assembly costs no accessibility, and retention approaching zero indicates complete occlusion of the metal by neighbouring clusters.

3.5 Monotonicity and its consequences

Passing from the isolated cluster to the assembly adds atoms and removes none. The union of inflated obstacle spheres therefore grows, the set of directions satisfying Equation 5 can only shrink, and the following bound holds for every metal atom and every probe radius,

$$\Omega_M^{\text{sheet}} \leq \Omega_M^{\text{cluster}}, \quad \text{hence} \quad 0 \leq \mathcal{R}(\Omega) \leq 1 \quad (12)$$

Two practical consequences follow. Any atom with zero accessibility in the isolated cluster retains zero accessibility in the assembly, which permits a large reduction in computational effort. More importantly, retention is a monotonically non-decreasing function of a . Any computed dependence that violates monotonicity signals a defective geometry rather than a physical effect, and Section 4 exploits the criterion.

4. Physical Validity of the Lattice Spacing

The hard-sphere model contains no repulsive term that prevents clusters from interpenetrating. Lattice spacings smaller than the contact separation may therefore be specified, and the calculation will return finite numbers that carry no physical meaning. A validity criterion is required.

The rigid contact separation, obtained as twice the maximum in-plane distance from the cluster centroid to any van der Waals surface, is 30.5 Angstrom for the present cluster. The minimum interatomic separation between the central cluster and its six nearest neighbours, evaluated as a function of a , provides a direct diagnostic and is given in Table 1.

a (Å)	gap relative to contact (Å)	minimum interatomic distance (Å)	assessment
22.5	-8.0	0.48	unphysical, atoms superposed

a (Å)	gap relative to contact (Å)	minimum interatomic distance (Å)	assessment
25.5	-5.0	0.64	unphysical, atoms superposed
28.5	-2.0	3.05	ligand shells interdigitated
30.5	0.0	4.38	ligand shells in contact
34.0	+3.5	7.46	clusters separated
40.0	+9.5	13.24	clusters well separated

Table 1. Diagnostic of geometric validity for the hexagonal assembly. Separations below approximately 2.5 Angstrom indicate that ligand arms of neighbouring clusters pass through the central cluster rather than packing against it.

Spacings at and below 25.5 Angstrom place non-bonded atoms within 0.7 Angstrom of one another, which is far inside any chemically meaningful contact distance. Accessibility values computed in that range are not merely inaccurate but non-monotonic in a , in direct violation of Equation 12, because whether a given pocket is occluded then depends on the accidental registry of interpenetrating ligands. Only spacings of 28.5 Angstrom and above are retained in the analysis that follows.

5. Results

5.1 Silver accessibility in the isolated cluster

Table 2 reports both descriptors for every silver atom that possesses non-zero accessibility at a probe radius of 1.4 Angstrom, corresponding to water. Forty of the forty-four silver atoms return exactly zero, meaning that no straight-line approach reaches them from any direction whatsoever. All twelve kernel atoms and all twenty dodecahedral atoms fall in that category, as does two thirds of the mount shell.

atom index	shell	distance from centroid (Å)	$\Omega / 4\pi$	SASA (Å ²)
Ag41	mount	6.38	0.0159	2.60
Ag34	mount	7.06	0.0091	1.15
Ag40	mount	6.28	0.0037	0.51
Ag39	mount	6.60	0.0008	0.13
remaining 40 Ag	kernel, dodecahedral, mount	2.29 – 7.08	0.0000	0.00
total over 44 Ag	—	—	0.0294	4.38

Table 2. Per-atom accessibility of silver in the isolated cluster at a probe radius of 1.4 Angstrom. Atom indices are zero-based and follow the input file ordering.

The accessible silver is confined entirely to the Ag₂(SR)₅ mount units, and a single atom, Ag41, accounts for 59 per cent of the total exposed silver area. Placed in context, the complete solvent-accessible surface of the cluster is 3302 square Angstrom, of which silver contributes 4.38 square Angstrom, or 0.13 per cent. The thirty sulfur atoms expose 29.5 square Angstrom, nearly seven times the silver value, so even the atoms directly bonded to the kernel are considerably better exposed than the kernel itself.

5.2 Silver accessibility in a hexagonal monolayer

Table 3 collects the summed descriptors for the isolated cluster and for the hexagonal assembly at each physically admissible spacing, together with the retention defined in Equation 11.

probe radius (Å)	representative species	$\Sigma\Omega/4\pi$, cluster	$\Sigma\Omega/4\pi$, $a = 28.5$ Å	$\Sigma\Omega/4\pi$, $a \geq 30.5$ Å	retention
1.20	H ₂	0.0525	0.0522	0.0525	0.993 – 1.000
1.40	H ₂ O	0.0294	0.0294	0.0294	1.000
1.65	CO ₂	0.0127	0.0127	0.0127	1.000
2.00	CH ₄ (approx.)	0.0025	0.0025	0.0025	1.000

Table 3. Summed accessible solid-angle fraction over all forty-four silver atoms, isolated cluster versus hexagonal assembly. Values at $a = 34.0$ and 40.0 Angstrom are identical to those at 30.5 Angstrom and are omitted.

The corresponding silver surface areas are 7.27, 4.38, 2.23 and 0.69 square Angstrom at probe radii of 1.20, 1.40, 1.65 and 2.00 Angstrom respectively, and are unchanged to three significant figures on passing from the isolated cluster to the assembly at every admissible spacing.

Assembly into the hexagonal arrangement therefore costs essentially no silver accessibility. The single measurable effect occurs for the smallest probe at the tightest admissible spacing, where retention falls to 0.993, a reduction of 0.7 per cent. The result is not an artefact of insufficient neighbour shells, since enlarging the image block leaves it unchanged, and it is consistent with Equation 12 in being monotonic in a .

The geometric explanation is direct. The four accessible silver atoms occupy shallow pockets on the mount units, and the surviving open directions are predominantly out of the assembly plane. Neighbouring clusters occupy in-plane positions where no accessible corridor existed in the isolated cluster, so their presence removes almost nothing. The organic shell has already occluded the metal before interparticle packing is considered.

5.3 Silver accessibility in three-dimensional superlattices

The insensitivity reported above is specific to a single layer and does not survive stacking. Table 4 gives the summed accessible solid-angle fraction for the three common close-packed stacking sequences at van der Waals contact, with a probe radius of 1.4 Angstrom. All three geometries have a minimum interatomic separation of 4.38 Angstrom and are therefore free of the interpenetration artefact discussed in Section 4.

stacking	description	$\Sigma\Omega/4\pi$	retention $\mathcal{R}(\Omega)$	retention $\mathcal{R}(A)$
monolayer	single hexagonal layer	0.0294	1.000	1.000
AAA	simple hexagonal	0.0264	0.900	1.000
ABCABC	cubic close packed	0.0229	0.779	1.000
ABAB	hexagonal close packed	0.0151	0.513	1.000

Table 4. Effect of stacking sequence on silver accessibility at $a = c = 30.5$ Angstrom and $rp = 1.4$ Angstrom. Retentions are given separately for the solid-angle descriptor of Equation 6 and the surface-area descriptor of Equation 8.

Stacking into three dimensions reduces the accessible solid angle by between ten and forty-nine per cent depending on the sequence. The hexagonal close packed arrangement is the most effective occluder because each cluster is capped directly by the threefold hollow of the adjacent layer, which lies along the out-of-plane direction that carries almost all of the surviving corridors. Simple hexagonal stacking places clusters in columns and leaves the intercolumn channels open, so it removes the least.

The dependence on the lattice parameters is monotonic, as required by Equation 12, and is given in Table 5 for the hexagonal close packed sequence.

a (Å)	c (Å)	minimum interatomic distance (Å)	$\Sigma\Omega/4\pi$	retention
30.5	26.0	4.38	0.0122	0.414
30.5	28.5	4.38	0.0137	0.466
30.5	30.5	4.38	0.0151	0.513
32.0	32.0	5.64	0.0186	0.631
34.0	34.0	7.46	0.0264	0.900

Table 5. Dependence of silver accessibility on lattice parameters for ABAB stacking at $r_p = 1.4$ Angstrom.

The probe dependence of the retention is weak over the chemically relevant range. For the hexagonal close packed sequence the retention is 0.540, 0.513 and 0.531 at probe radii of 1.20, 1.40 and 1.65 Angstrom respectively, and for the cubic close packed sequence it is 0.784, 0.779 and 0.791. The occlusion is therefore a property of the assembly geometry rather than an accident of one probe size, which is the expected behaviour when the effect arises from blocking of extended corridors rather than from marginal contacts.

5.4 Divergence of the two descriptors

The surface-area retention in Table 4 is unity for every stacking, while the solid-angle retention falls to 0.513. The two descriptors disagree completely, and the disagreement is informative rather than contradictory.

The reason is a difference in locality. Equation 8 evaluates burial on a sphere of radius $R_M + r_p$ centred on the metal atom, a surface that lies within roughly 3.1 Angstrom of the silver nucleus and therefore deep inside the ligand shell. Atoms of neighbouring clusters never approach within 4.38 Angstrom of any atom of the central cluster, so they cannot intersect that sphere and cannot alter the area. Equation 6, by contrast, tests the entire path from the contact radius outward, and neighbouring clusters intercept those paths several Angstrom beyond the ligand surface.

A practical consequence follows for the wider literature. Solvent-accessible surface area, which is the descriptor most commonly reported for protected clusters, is structurally blind to assembly effects by construction and will report no change on aggregation regardless of how completely the metal is enclosed. Any assessment of metal accessibility in an assembled material should therefore use a path-based criterion, and reporting surface area alone would lead to the incorrect conclusion that assembly is inconsequential.

5.5 Probe-size selectivity

The accessibility descriptors fall by a factor of twenty-one between probe radii of 1.20 and 2.00 Angstrom, and the number of silver atoms retaining any exposure falls from four to two. At 1.20 Angstrom the exposed set is Ag34, Ag39, Ag40 and Ag41. At 1.65 Angstrom the set reduces to Ag34, Ag40 and Ag41, with Ag40 retaining only 0.09 square Angstrom. At 2.00 Angstrom only Ag41 remains meaningfully exposed, at 0.69 square Angstrom.

The steep dependence establishes that the accessible corridors are narrow rather than merely shallow, and it implies pronounced size selectivity. A small molecule such as dihydrogen or water may reach several hundred silver atoms, whereas a species of the size of methane is restricted to a single site on

the entire cluster. Reporting either descriptor without the associated probe radius is therefore meaningless.

6. Discussion and Limitations

Three limitations bear on the interpretation of the numbers above.

First, the analysis is purely geometric and static. A rigid hard-sphere model measures whether a corridor exists, not whether a molecule will occupy it. Binding energetics, ligand polarisation and solvent structuring are absent, and a site with a favourable corridor may still bind weakly.

Second, the ligand shell is treated as frozen at a single geometry. Carboxylate and nitro substituted aromatic arms reorient on picosecond timescales, and transient openings will raise the time-averaged accessibility above the static value. Averaging the descriptors over molecular dynamics snapshots would give a fairer estimate for the isolated cluster. Ligand mobility would also allow neighbouring shells in the assembly to interdigitate properly rather than merely abut, which is the one mechanism by which packing might reduce accessibility more than the present static treatment indicates.

Third, the assembly is idealised. All clusters share one orientation, the lattice is defect free, and the spacing is imposed rather than relaxed. An experimental centre-to-centre distance from diffraction or scattering data, or a spacing obtained from relaxation, should replace the rigid contact estimate before the retention is quoted as a property of a real assembly.

A fourth point concerns what the calculations do and do not determine. The stacking sequence and lattice parameters of the assembled material have not been established experimentally in the present work, and the retention varies from 0.90 to 0.41 across the geometries examined. The range is reported rather than a single value, and a specific figure should not be quoted until the structure is known. The qualitative conclusion, that three-dimensional stacking substantially reduces silver accessibility while a single layer does not, holds across the entire range and does not depend on that choice.

Subject to those limitations, two conclusions follow. Within a single hexagonal layer the accessibility of silver is limited by the thiolate shell rather than by interparticle packing, since neighbouring clusters occupy in-plane positions from which no approach corridor existed in the isolated cluster. On stacking into three dimensions the surviving corridors, which are predominantly out of plane, are intercepted by clusters in adjacent layers, and between ten and forty-nine per cent of the accessible solid angle is lost. Dimensionality and stacking sequence, rather than in-plane density, are therefore the structural variables that govern metal accessibility in assemblies of this cluster.

7. Computational Details

Van der Waals radii were taken from the Bondi compilation, with 1.72 Angstrom for silver, 1.80 for sulfur, 1.70 for carbon, 1.55 for nitrogen, 1.52 for oxygen and 1.20 for hydrogen. Solid-angle integration used 40000 Fibonacci directions per atom. Comparison against 20000 directions changes the summed descriptor by less than 0.5 per cent, and the residual quadrature uncertainty on an individual value of 0.016 is estimated at 0.0006. Surface areas used 4802 quadrature points per atom. Obstacle atoms beyond 22 Angstrom of the site were neglected, a cut-off that exceeds the largest sum of inflated radii by more than a factor of five. The hexagonal monolayer used a five by five block of images with the assembly normal along the laboratory z axis. Three-dimensional superlattices used a seven by seven in-plane block with one layer above and one below. Enlarging the in-plane block to nine by nine or adding a second layer in each direction changes the summed descriptor by less than

one part in ten thousand. All calculations were performed with an in-house implementation requiring only NumPy.

The bound in Equation 12 was used as a runtime correctness check throughout, and no computed retention exceeded unity at any admissible spacing.